

OS Types

Batch OS
 Multiprogramming OS
 Multitasking OS
 Time Sharing OS
 Multiprocessing OS
 Distributed OS
 Network OS
 Mobile OS
 Real time OS (Hard/Soft)

Influences
Choice of

OS Structures

- Simple / Monolithic
- Layered Approach
- Microkernel
- Modular
- Hybrid
- Virtual Machine Structure

Determines
Implementation
of

Operating System

performs

Based on
Structure

provide
Interface
through

Invokes

System Calls

- process control → fork(), exec(), write(), wait()
- File management → open(), read(), write(), close()
- Device Management → ioctl(), read(), write()
- Information → get pid(), time(), uname()
- Communication → pipe(), socket(), shmget()
- protection → chmod(), access()

Decide
Behavior
of

Support
Working
of

→ Direct relation

----> Influences

----> Depends on

----> supports

----> invoke

OS Functions

- Process management
- memory management
- File management
- Device Management
- I/O Management
- CPU Scheduling
- Security & protection
- User interface (CLI, GUI)
- Error Detection
- Resource Allocation

Key Idea

Operating System acts as an intermediary between user programs and hardware resources and manages all system activities.